

Customer Shopping Behavior Analysis

A review of 3,900 customer transactions to understand who is buying, what they're buying, and whether current promotions and loyalty programs are paying off.

Prepared for: Retail Operations & Marketing Stakeholders

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Data period: Full transaction extract, 3,900 orders

Tools used: Python (cleaning), PostgreSQL (analysis), Power BI (visualization)

Executive Summary

\$233,081 TOTAL REVENUE	3,900 ORDERS ANALYZED	\$59.76 AVG. ORDER VALUE	3.75 / 5 AVG. REVIEW RATING
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THE ASK

Review a year's worth of customer shopping data to understand purchasing patterns. Who is buying, what they're buying, where, and how. Also assess whether current promotions, discounts, and the subscription program are actually working in the business's favor.

WHAT WE DID

- Cleaned and standardized 3,900 customer transaction records (column names, data types, category labels).
- Investigated 37 orders with a missing review score and made a documented decision on how to handle them rather than guessing.
- Loaded the clean dataset into a PostgreSQL database and wrote 17 business questions as SQL queries covering sales, customers, products, and payments.
- Built a four-page Power BI dashboard so results are easy to explore and re-check by category, location, season, and customer group.

WHAT WE FOUND

- **Clothing drives the business.** It accounts for 45% of all revenue (\$104K), roughly double the next category (Accessories, \$74K).
- **Customers aged 51+ spend the most.** The 51-65 and 36-50 groups together account for \$133K, well over half of total revenue.
- **Discounts and promo codes are not increasing basket size.** Orders placed with a discount or promo code average \$59.28, essentially the same as \$60.13 for orders without one.
- **The subscription program shows almost no revenue lift.** Subscribers spend about the same per order as non-subscribers (\$59.49 vs. \$59.87).
- No location, season, or shipping method stands out as dramatically over- or under-performing. Sales are fairly evenly spread, which is itself a useful finding.

WHAT IT MEANS FOR THE BUSINESS

Revenue is healthy and concentrated in a few clear areas (Clothing, older age groups), which makes targeting easier. But two cost-generating programs, discounts/promo codes and the subscription tier, are not currently translating into bigger purchases. That's worth a closer look before renewing budget for either, and it's the main actionable finding in this report.

The Question We Were Asked to Answer

Original request

Leadership wanted a clear picture of customer shopping behavior across the store: which products and categories perform best, which customers are most valuable, and, importantly, whether the discounts, promo codes, and subscription program that the business pays for are actually changing how much customers spend.

What we were asked to do

Take the raw point-of-sale export, clean it into a trustworthy dataset, and answer a set of concrete business questions: top-selling products and categories, best customer segments, best-performing locations and seasons, and, the question with the most direct financial impact, whether promotional spend is working.

Why this matters

Every dollar spent on discounts, promo codes, and subscription perks is a dollar the business expects to earn back through larger or more frequent orders. Without measuring that, it's a guess. This report replaces the guess with evidence.

What We Did

1. Cleaned the data

The raw export had a few issues that needed handling before any numbers could be trusted:

- Column names were inconsistent (spacing, capitalization) and were standardized.
- 37 orders (about 1% of the dataset) were missing a review score. We checked whether these were random or followed a pattern. They weren't random. **All 37 belonged to male customers who had a discount and a promo code applied**, suggesting a gap earlier in the customer journey (e.g., the review step wasn't reached) rather than a data-entry error. We flagged this rather than filling it with a guessed value.
- Yes/No fields (subscription, discount, promo code) were converted to a consistent True/False format so they could be counted and compared reliably.
- Ages were grouped into five bands (18-25, 26-35, 36-50, 51-65, 65+) to make customer comparisons easier to read.
- We checked for duplicate customers and outlier values (age, order size). None were found. The dataset is clean and internally consistent.

Data note: No sales dates were available in this export, so this report describes overall patterns across the full extract rather than trends over time (e.g., month-over-month growth).

2. Answered the business questions in SQL

The cleaned data was loaded into PostgreSQL, where we wrote 17 queries covering sales by category, customer segment, product, location, season, payment method, and shipping preference, including which discount and loyalty behaviors correlate with spend.

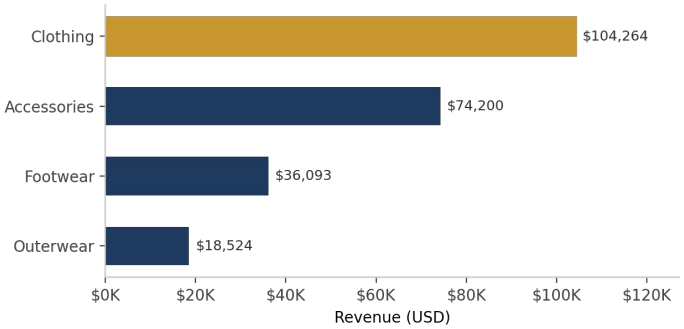
3. Built a dashboard

Results were visualized in a four-page Power BI dashboard (Sales Overview, Customer Insights, Product Overview, Payments & Logistics) so the findings can be explored interactively rather than read only as a static report.

What We Found

Clothing is the clear revenue leader

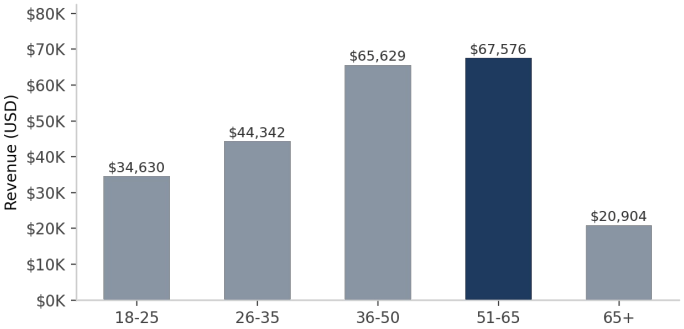
Clothing generated \$104,264, which is 45% of all revenue, nearly double Accessories, the second-largest category. Footwear and Outerwear trail well behind.



Total revenue by product category (USD)

Older customers spend the most

The 51-65 age group leads in total spend (\$67,576), closely followed by the 36-50 group (\$65,629). Together, customers over 36 account for the majority of revenue, while the 18-25 group contributes the least.

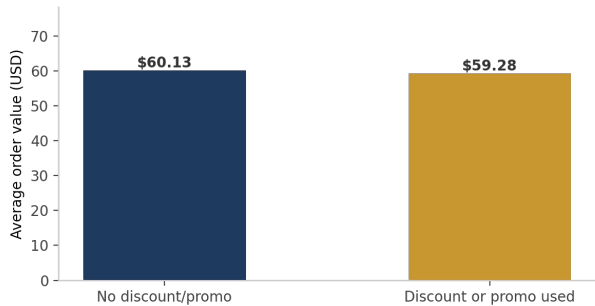


Total revenue by customer age group (USD)

What We Found: Promotions & Loyalty

Discounts and promo codes aren't increasing order size

This is the most important finding in the report. Orders that used a discount or promo code averaged **\$59.28**, slightly *lower*, not higher, than the **\$60.13** average for orders with no discount at all. In other words, customers who get a discount aren't spending more to earn it; the discount is simply reducing margin on a sale that likely would have happened anyway.



Average order value: with vs. without a discount/promo code

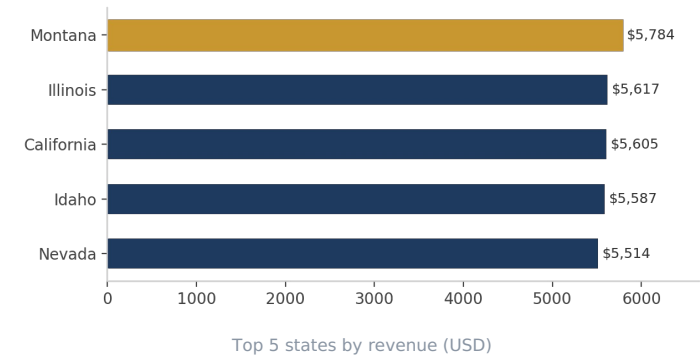
The subscription program shows the same pattern

Subscribers spend \$59.49 per order on average versus \$59.87 for non-subscribers. Essentially, no difference. Subscribers do have slightly more prior purchases on record (26.1 vs. 25.1), so the program may be retaining customers, but it is not currently driving bigger baskets.

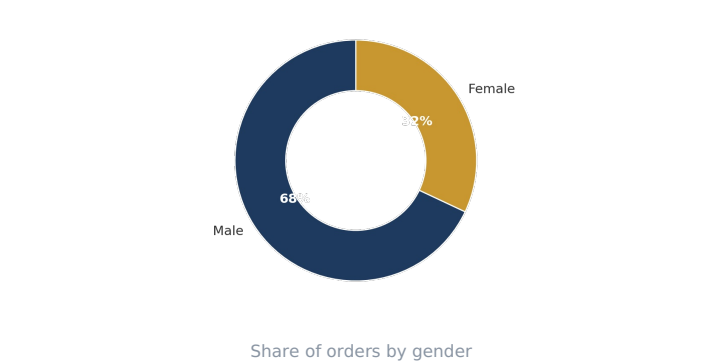
Why this matters financially: Discounts and subscription perks cost the business money on every order. Right now, that cost isn't showing up as extra revenue per order. This doesn't mean the programs should be cancelled. They may help with retention or acquisition in ways this dataset can't measure, but the assumption that "discounts increase basket size" doesn't hold up in this data.

What We Found: Products & Operations

Top locations by revenue



Customer base skews male



Shipping preference is evenly spread

No single shipping method dominates. Free Shipping is the most-used option, but the six methods are all within a narrow range of each other (627–675 orders). There's no strong signal that customers are avoiding paid shipping or overwhelmingly favoring speed.



Other consistent patterns

Question	Finding
Best-selling size (across categories)	Medium is #1 in every category; Large is the runner-up
Highest-rated category	Footwear (3.79 avg). Categories are close, none stands out as a quality problem
Most-used payment method	Credit Card, PayPal, and Cash are used almost equally. No single method dominates
Season with highest revenue	Fall (\$60,018), but all four seasons are within ~7% of each other

What This Means for the Business

The headline takeaway

The business has a healthy, well-distributed revenue base led by Clothing and by customers over 36. That's good news and a clear place to focus marketing spend. But the promotional tools currently in place (discounts, promo codes, and the subscription program) are not measurably increasing how much each customer spends. If the goal of those programs was to grow order size, **the data says they haven't**.

Did the changes lead to more customer spending?

Based on this snapshot, no, not on a per-order basis. Average order value is essentially flat regardless of whether a discount, promo code, or subscription was involved. If anything, discounted orders trend very slightly lower.

What could be done differently

- 1 Re-test the discount strategy.** Consider a controlled A/B test. Offer the same discount to only half of a customer segment and compare basket size directly, rather than relying on who happens to use a promo code.
- 2 Investigate the subscription program's real value.** If it isn't growing order size, check whether it's improving purchase frequency or retention instead. That may be its actual benefit, and worth measuring directly.
- 3 Lean into what's working.** Clothing and the 36+ age groups are the strongest performers. Consider concentrating promotional budget there instead of spreading it evenly across all customers.
- 4 Add order dates in future exports.** This dataset had no transaction dates, which ruled out trend analysis (growth over time, seasonality by month). That's a quick fix for the next data pull and would unlock a much deeper follow-up study.
- 5 Follow up on the missing-review pattern.** All 37 orders missing a review score shared the same profile (male, discount + promo used). Worth checking if there's a step in the post-purchase flow that's being skipped for that group.

Full SQL queries, the cleaning notebook, and the interactive Power BI dashboard are available in the project repository. This report reflects a single historical extract of 3,900 transactions and does not include order dates, so findings describe overall patterns rather than trends over time.